

TrustEpisode: Auditable Evidence Contracts for AI-Assisted EDR/NDR Investigation

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Abstract

AI-assisted security investigation can turn sparse tool outputs into confident narratives, but arrival order, collector failure, and late evidence can silently change the facts exposed to an analyst or agent. We present TRUSTEPISODE, a deterministic and auditable evidence contract for EDR/NDR episode reconstruction. Episode Synthesis is the sole membership owner; revisions and terminal audit outcomes are append-only; source health remains distinct from “no detection,” and unsupported probabilities remain null. Downstream AI is a read-only consumer of versioned evidence rather than an authority over case facts.

On a sealed live CALDERA cohort (70 M1–M7 malicious and 40 benign runs), sliding sessionization raises mean coverage from 0.807 to 0.821 versus fixed bins, while fair sliding baselines do not establish superiority: EB4 matches EB2 on all action-coverage pairs (0/70/0 W/T/L), but trails EB1 on one concurrent-benign M7 run (0/69/1; mean $\Delta = -0.014$, 95% CI $[-0.043, 0.000]$). Canonical ordered replay is byte-identical on all 60 M1–M6 runs; raw arrival order is not robust without re-canonicalization. 18/18 audit fixtures cover all seven terminal failure codes, and an A–G evidence-loss taxonomy localizes M2’s repeatable gap to D_token_mapping_failure. A deployed case trace preserves one Suricata and two Wazuh events plus ATT&CK support references in the analyst surface. All 7/7 synthetic downstream-mutation cards pass: the valid canonical control is accepted and six injected or tampered records are rejected. A once-run held-out 20-card local Qwen2.5-7B pilot produces schema-shaped JSON on all cards, but only 15/20 exact answers, 15/20 exact citation sets, and 9/20 fully exact responses. A locked same-card replication on guacamole-ai’s default Gemma3-1B drops to 7/20 exact answers, 0/20 exact citations, and 0/20 fully exact responses. We therefore make no general LLM-accuracy or fitted-calibration claim: model behavior is size-sensitive and untrusted, while typed evidence makes its failures reviewable.

CCS Concepts

• **Security and privacy** → **Intrusion/anomaly detection and malware mitigation**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

AI-assisted security operations, attack episode reconstruction, EDR, NDR, replayable audit, evidence provenance

1 Introduction

AI-assisted SOC systems increasingly use language-model agents to gather tool outputs, correlate alerts, and draft incident narratives [5, 21, 25]. Their output can be no more reviewable than the evidence interface below it: if arrival order changes episode membership, collector failure is encoded as benign silence, or a late event rewrites

history, a fluent explanation may cite a case state that cannot be reproduced.

Security operations and digital-forensic investigations therefore need more than alert clustering. Multi-step attacks appear as dispersed EDR and NDR observations that become material when linked by time, entity, session, and process relations [1, 11, 15]. Investigators need a contract that fixes who owns episode membership, how late evidence is admitted, how collector health differs from “no detection,” and how every revision leaves an independently verifiable terminal outcome [3, 4, 16].

Prior alert-correlation and provenance systems motivate multi-stage context using rules, graphs, and learned representations [6, 8–10, 12, 20]. Those lines do not remove the need for fixed ownership of membership, immutable revision history, and replayable audit semantics under incomplete observability. This submission evaluates TRUSTEPISODE as a *deterministic, auditable reconstruction contract* for EDR/NDR evidence and as a typed, read-only boundary for downstream analyst or AI reasoning. The downstream model is deliberately outside the trusted computing base: it may summarize versioned AuditRecords but cannot own membership, infer source health, overwrite history, or convert unsupported evidence into a numeric probability. A compact scorer and calibrator remain optional schema interfaces and are not empirically validated here. Thus, our AI-security claim concerns *evidence truthfulness and containment*, not broad LLM effectiveness. We additionally report a small, held-out Qwen pilot and a post-result locked replication on the deployment-default Gemma model to measure extraction, citation, and abstention over the typed record. They are boundary stress tests rather than a preregistered model comparison.

Research questions.

- RQ1** Under identical sealed EDR/NDR evidence, does TrustEpisode reproduce byte-identical membership, revision history, late-evidence records, and digests on all 60 malicious runs?
- RQ2** Do open, finalized, late-successor, closed, beyond-horizon, duplicate, and all seven AssemblyFailureRecord codes yield explicit immutable audit outcomes without rewriting earlier history?
- RQ3** When a required action is absent from a reconstructed episode, can the pipeline localize loss across the closed A–G taxonomy, and is that taxonomy instantiated on sealed live evidence?
- RQ4** Which relation, hub, merge, and late-revision mechanisms does the live M1–M7 cohort exercise, and which require targeted conformance cards?
- RQ5** What portion of observed formation gain comes from replacing fixed bins with sliding sessionization, and does relation-aware logic provide additional measurable benefit on the sealed cohort?
- RQ6** Do closed schemas and digest bindings fail closed when a downstream consumer injects asserted facts, labels, or unsupported

probability; removes provenance; or tampers with revision and evidence digests, and how reliably do two local models follow the resulting read-only extraction, citation, and abstention interface?

These questions evaluate the evidence substrate. The deployed interface trace reported later checks end-to-end materialization but is not counted as an effectiveness trial, and its displayed operational confidence is not the unevaluated calibrated probability field defined by the contract.

Contributions.

- (1) an agent-facing, single-owner immutable reconstruction contract: Episode Synthesis alone owns membership; late evidence yields successors or LateEvidenceRecords; deadlines do not extend; terminal history is append-only; downstream AI remains read-only; a seven-card synthetic mutation suite tests canonical-record injection and tampering; a 20-card Qwen pilot plus a locked same-card Gemma replication expose size-sensitive extraction, citation, and abstention failures;
- (2) replayable forensic audit outcomes with exactly one terminal AssemblyOutcome per revision, canonical references, SHA-256 digests, and a complete seven-code failure matrix;
- (3) a stage-localized A–G evidence-loss taxonomy, conformance-tested across classes and instantiated on sealed M2 as D_token_mapping_failure;
- (4) a fair negative-result evaluation: legacy fixed bins underperform sliding sessionization; EB4 matches EB2 on all 70 action-coverage pairs but trails EB1 on one concurrent-benign M7 run (the paired CI includes zero); advanced guards require targeted cards not mixed into primary effectiveness;
- (5) a determinism boundary on all 60 sealed malicious runs: canonical ordered replay and duplicate retries are stable; raw delivery-order permutations are not order-robust without re-canonicalization; formation is single-writer per run;
- (6) a contract-ablation comparison (AB0–AB3) isolating which forensic failure modes mutable, noncanonical, and silent-failure stores admit relative to TrustEpisode; and
- (7) a deployed, read-only case-materialization trace showing that source-separated Suricata and Wazuh events, ATT&CK technique links, and evidence references reach the analyst surface without being promoted to a calibration or LLM-effectiveness claim.

2 Related Work

2.1 Alert Correlation and Attack Reconstruction

Classical alert correlation reconstructs multi-step scenarios from intrusion alerts by temporal and causal linking [12]. Provenance-based detectors such as HOLMES, UNICORN, KAIROS, NODLINK, and CAPTAIN recover APT context from system graphs and alerts [6, 8–10, 20]. Those systems motivate episode construction; they do not by themselves fix single-owner membership, late-data revision policy, or append-only terminal audit outcomes for mixed EDR/NDR streams.

2.2 Digital Forensic Event Reconstruction

Timeline-based event reconstruction is a core digital-forensic technique for inferring past activity from heterogeneous artifacts [1, 4]. Recent work formalizes timestamp interpretation via time anchors and clock-skew anomalies [16] and surveys challenges beyond mere data volume [1]. Process models and filesystem analysis emphasize documenting what was examined, what was unavailable, and how conclusions depend on incomplete traces [3, 11]. TrustEpisode imports these audit requirements into *online* EDR/NDR episode reconstruction: immutable revisions, explicit late stores, and digestable terminal outcomes under delayed delivery.

2.3 Reproducibility, Tool Validation, and Provenance

Forensic tool and method validation requires documented procedures, known datasets, and repeatable results before casework [2, 14, 18]. Chain-of-custody and provenance discipline demand that investigators show integrity of examined objects and distinguish observed from unavailable evidence [4, 11]. TrustEpisode’s contribution is not a higher detector F1; it is a single-owner revision contract, canonical replay boundary, and explicit terminal failure semantics that make reconstruction outcomes auditable under the same reproducibility expectations.

2.4 Incomplete Telemetry and Evidence-Loss Attribution

EDR and NDR fail differently; missing modality, parser blind spots, and collector health can erase or distort attack steps without implying that correlation “missed” an already-normalized event [17, 23]. Reconstruction contracts must therefore separate sensor observability, normalization/token mapping, formation membership, and offline attribution when diagnosing incomplete episodes.

2.5 Evidence-Grounded and Contained AI Investigation

Tool-using language-model systems interleave reasoning with external observations [25]. In security operations, CORTEX coordinates specialized agents over enterprise evidence, while Holmes packages network evidence into auditable anchors for an LLM investigator [5, 21]. These systems motivate evidence-grounded AI assistance, but grounding also needs an upstream contract: citation presence does not by itself establish faithfulness [7, 19], and agent execution requires architectural containment rather than prompt-only trust [24]. TRUSTEPISODE addresses the complementary boundary below the agent. It fixes which EDR/NDR objects exist, which producer owns them, how missingness is represented, and which immutable references a later narrative may cite. We do not compare LLMs or claim better autonomous triage.

2.6 Positioning Summary

Table 1 contrasts representative lines along forensic-contract properties. Unknown cells are marked *Not reported*; we do not infer undocumented product behavior. TRUSTEPISODE’s novelty is the audit lifecycle, evidence-loss localization, and read-only agent-facing evidence boundary, not a new graph algorithm or classifier.

Table 1: Related-work comparison on forensic reconstruction contract properties (Not reported = not documented in the cited source).

Work	EDR+NDR	Imm. rev.	Late evid.	Canon. replay	Term. fail	Src health	Loss loc.	Live
Alert correlation [12]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
HOLMES [10]	Partial	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
UNICORN [8]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
KAIROS [6]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Not reported	Yes
Timeline SoK [1]	Survey	Survey	Survey	Survey	Survey	Survey	Survey	Survey
Time anchors [16]	No	Not reported	Not reported	Not reported	Not reported	Not reported	Clock skew	Yes
TRUSTEPISODE (this)	Yes	Yes	Yes	Yes	Yes	Yes	A-G	Yes

2.7 Scope Distinction

TRUSTEPISODE does not claim a new graph detector, AI ranking effectiveness, attribution beyond observable EDR/NDR, enterprise impact estimation, or autonomous response. The evaluated contribution is a deterministic forensic reconstruction and audit contract for sealed EDR/NDR evidence under late arrival and incomplete observability. Its AISec relevance is architectural: AI assistance is constrained to read and cite versioned case facts rather than silently redefining them.

3 TrustEpisode Reconstruction Contract

TRUSTEPISODE converts EDR/NDR telemetry into deterministic episode revisions and version-scoped forensic audit records. Evidence Fabric and Health Monitor own events and source health; Episode Synthesis alone owns membership; optional Feature Extractor / Scorer / Calibrator fields remain schema interfaces without fitted artifacts in this submission. For each revision, the AuditRecord Assembler emits exactly one terminal AssemblyOutcome: a canonical AuditRecord or an immutable AssemblyFailureRecord. Ground truth is joined only offline. Figure 1 shows these boundaries.

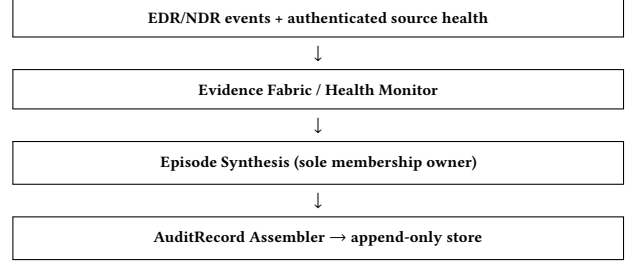
3.1 Investigation and Threat Model

Assumptions. Raw pointers and the sealed evidence store are integrity-checkable; Health Monitor coverage messages have authenticated sources; attackers may induce missing events, delayed delivery, and duplicate retries; attackers cannot arbitrarily forge signed health state; runtime software may fault; the offline evaluator is isolated from online formation.

Non-assumptions. We do *not* assume complete telemetry, that collector silence equals downtime, that all attacks are observable, that ground truth is available online, that model scores are reliable, or that raw arrival order is fixed.

Downstream-agent adversary. An AI assistant may be mistaken, prompt-injected, or compromised: it may omit evidence, misquote a record, or recommend an unsafe action. It is outside the trusted computing base and has no capability to write canonical evidence objects. The contract does not make its narrative true; it makes any cited case claim checkable against an immutable revision and prevents the narrative from silently becoming the forensic record.

Investigator workflow. (1) import sealed EDR/NDR; (2) form EpisodeRevisions; (3) inspect the terminal audit outcome; (4) check coverage and provenance; (5) accept a late successor or LateEvidenceRecord; (6) if a required step is missing, run the A-G waterfall; (7) replay and verify digests; (8) report observed, unavailable, and unknown separately.



Offline only: labels, matching, evaluation.

Figure 1: Label-blind online reconstruction and offline evaluation boundary.

3.2 Evidence and Reconstruction Model

Online objects are NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, AuditRecord, and AssemblyFailureRecord. Offline labels never enter online schemas. Table 2 assigns every object to one producer.

3.3 Independent Evidence Health and Observability

Event silence is not an outage. “No detection” is not “unavailable.” SourceCoverage carries authenticated health, parser validity, and an observed source mask. Assemblers consume exact coverage references; they do not infer health from empty event windows.

3.4 Deterministic Episode Formation

Formation links events using process-causal, authenticated-session, endpoint-connection, and specific-entity relations under a 300 s maximum inter-event gap and 3600 s maximum episode span. Tie-breaks are deterministic (relation precedence, relation count, Δt , start time, episode ID). Hub suppression, parent merge, and exact event-time ordering are part of the frozen policy. Formation never reads labels, receipts, or scenario metadata.

3.5 Immutable Revision Lifecycle

Watermark lag finalizes open episodes. Late evidence inside a fixed 900 s revision horizon may create an immutable successor revision; beyond-horizon or closed-lineage evidence produces a LateEvidenceRecord. Earlier revision digests are never rewritten. Deadlines are not extended by late arrivals.

3.6 Terminal Forensic Audit Contract

At revision.watermark+120 s the assembler inserts exactly one terminal outcome under (*episode_id*, *revision*): AuditRecord or AssemblyFailureRecord with a closed failure code set (schema, key,

Table 2: Single-owner online object contract.

Object	Sole producer	Consumers	Forbidden behavior
NormalizedEvent	Evidence Fabric	Episode Synthesis	label, scenario, or health fields
SourceCoverage	Health Monitor	Scope Resolver; Sufficiency; Assembler	infer outage from event silence
EpisodeRevision / LateEvidenceRecord	Episode Synthesis	Feature; Sufficiency; Assembler / late store	rewrite a prior revision; admit beyond-horizon into membership
FeatureVector $[D, C, m]$	Feature Extractor	Scorer; Assembler	labels or inferred health
Nullable P	Calibrator (design)	Assembler	numeric unsupported output
Structured U	Sufficiency Evaluator	Assembler	scalarization that rewrites P
AssemblyOutcome	AuditRecord Assembler	Append-only store; replay	both success and failure; update/delete

Deterministic episode and revision lifecycle

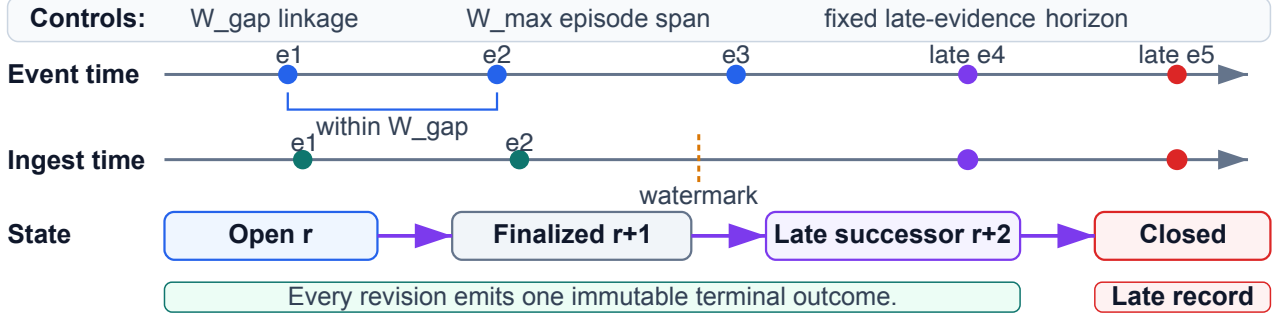


Figure 2: Episode revision lifecycle: open $\rightarrow$ finalized $\rightarrow$ late successor or closed; beyond-horizon evidence enters LateEvidenceRecord without rewriting earlier membership.

duplicate, digest, version, missing-input, timeout). Success cannot follow terminal failure. References are canonically ordered; RFC 8785/JCS serialization and SHA-256 digests bind versions. The store is append-only.

3.7 Agent-Facing Evidence Boundary

Downstream analyst views and AI assistants are consumers, never producers, of canonical case facts. They receive a revision key, terminal AssemblyOutcome, ordered evidence references, SourceCoverage references, and the independent sufficiency profile. A generated explanation or recommended action is a noncanonical view: it cannot write EpisodeRevision, change P or U , or replace an AssemblyFailureRecord with success. If required inputs are missing, the only canonical output is the failure record; an agent is not permitted to fill the gap from conversational context. This boundary makes a later narrative falsifiable against immutable source references without treating the language model as part of the forensic root of trust. Section 4.13 tests closed-schema, support-coupling, provenance, and digest guards. Section 4.14 then treats the model as an untrusted reader and scores exact values, abstention, and pointer citations; service-identity authorization remains a deployment assumption rather than an evaluated result.

3.8 Evidence-Loss Taxonomy

When a required action token is absent from a reconstructed episode, diagnosis uses the following closed classes (Table 3):

Forbidden inference: treating $F1 = 1.0$ as exact-complete step recovery; treating missing tokens as formation failures without a waterfall.

Table 3: Evidence-loss taxonomy for incomplete reconstruction.

Class	Decision rule (summary)
A_execution_failure	Action receipt/predicate shows non-success
B_sensor_observability_failure	Receipt success but raw EDR/NDR lacks identifiable activity
C_parser_or_normalization_failure	Raw present; NormalizedEvent absent or rejected
D_token_mapping_failure	NormalizedEvent present under wrong/missing required token
E_formation_membership_failure	Correct token exists but not retained in matched episode
F_offline_attribution_failure	Episode retains evidence; matcher fails to credit token
G_unknown_or_incomplete_trace	Required supporting records missing for A-F

3.9 Scoring and Probability Extension (Design Only)

Schema fields may retain an optional score interface: version-bound, nullable P outside support, and U that must not rewrite P . This submission reports no fitted coefficients, Brier/NLL, or held-out ranking. Scoring and calibration are not empirical claims here.

3.10 Boundary

The *evaluated* contribution is deterministic episode reconstruction and replayable audit semantics, including evidence-loss localization and a deployed read-only case-materialization trace. Manifest-scoped probability is design-only. LLM generation quality, CTI enrichment, impact scoring, autonomous response, and OT/ICS generalization are excluded.

4 Evaluation

4.1 Evidence Classes and Cohorts

- **RQ1** reproducibility — sealed offline replay on all 60 primary malicious runs.

- **RQ2** lifecycle auditability — A1–A12 synthetic audit conformance covering seven failure codes + late/beyond-horizon cards.
- **RQ3** evidence-loss localization — A–G taxonomy conformance + sealed M2 waterfall.
- **RQ4** mechanism coverage — live M1–M7 + synthetic T1–T8, with M7 as boundary stress.
- **RQ5** baseline interpretation — sealed fair EB0–EB4 comparison on M1–M7.
- **RQ6** downstream containment and reading — synthetic AI0–AI6 canonical-record validation plus two once-run 20-card local-model grounding audits.

Evidence classes used below are disjoint: (i) primary live empirical (M1–M7 formation cells); (ii) sealed offline replay on those primary seals; (iii) synthetic conformance (A1–A12, T1–T8, AB0–AB3, AI0–AI6, A–G fixtures); (iv) two once-run synthetic held-out LLM audits; (v) unit/integration boundary fixtures; (vi) supplemental partition artifacts (L1–L4, wall-clock) that are *not* mixed into primary estimators. Synthetic cards are never mixed into primary live effectiveness. Sealed Table 5 primary cells were not overwritten.

4.2 Primary Live M1–M7 Cohort

CALDERA v5.3.0 executes frozen Bash abilities on two isolated Ubuntu 22.04 targets with an EDR wrapper and inline Suricata NDR. Online schemas reject labels and receipts. The primary sealed cohort has 70 malicious runs (M1–M7 $\times$ two modes $\times$ five) and 40 standalone benign runs. Fair sliding baselines and paired comparisons are computed on this M1–M7 set.

4.3 Ground Truth and Matching

Offline matching uses temporal IoU ≥ 0.10 , entity Jaccard ≥ 0.50 , overlapping required action tokens, and weight

$$w = 0.5A_{cov} + 0.3J_{entity} + 0.2IoU_{time}; \quad w \geq \theta_{match} = 0.50. \quad (1)$$

Exact-complete recovery requires $A_{cov} = 1$. Online builders never read evaluation fields.

4.4 Fair Reconstruction Baselines

- **EB0_FIXED_BIN_300s** (legacy): fixed bins; boundary reference only.
- **EB1_SLIDING_TIME**: same $W_{gap} = 300$ s, $W_{max} = 3600$ s; time only.
- **EB2_SLIDING_ENTITY**: same lifecycle; entity overlap only.
- **EB3_RELATION_NO_ADVANCED_GUARDS**: relations on; hub/merge/late off.
- **EB4_TRUSTEPIISODE**: full production policy.

Primary comparison is EB1–EB4 on identical sealed streams, aliases, matching, and ground truth. Configuration digests are sealed in `results/derived/fair_baselines.json`.

4.5 Primary Formation Results and Fair Negative Result

Table 5 reports production-formation cells for the primary live M1–M7 cohort ($n = 5$ per cell). M7 is a boundary-stress family: the final action occurs at 3900 s, beyond the frozen $W_{max} = 3600$ s span, so

zero coverage/F1 is contract-conformant rather than a collection failure.

Relative to legacy EB0, sliding policies raise coverage (0.807→0.821) and exact-complete recovery (0.671→0.714) on the sealed primary cohort. Fixed bins cut multi-minute chains at arbitrary boundaries; sliding gap/sessionization preserves those chains under the same span cap.

The fair sliding baselines are not identical. EB4 matches EB2 on all action-coverage pairs (0/70/0 W/T/L). Against EB1, action coverage and exact-complete recovery have 0/69/1 W/T/L, mean $\Delta = -0.014$, and 95% CI $[-0.043, 0.000]$; episode-F1 has mean $\Delta = -0.010$ and 95% CI $[-0.029, 0]$. The single loss is the concurrent-benign M7 run `run-ee27d58be2db032c44d0774d9358c4b1`: EB1 recovers its truth group while EB4 does not. We therefore report the EB4–EB1 result as *not statistically resolved*, not “no observed difference,” equivalence, or relation-aware superiority. EB3 shows the same one-run action-coverage pattern.

Why sliding-time already suffices on M1–M7. The sealed primary cohort does not make advanced relations the unique linkage path: EB4 and EB2 match on all 70 action-coverage pairs, and EB4 matches EB1 on 69. On the remaining concurrent-benign M7 run, EB1 and EB3 recover a truth group that EB4 and EB2 do not. Thus the advanced-guard policy incurs one observed boundary-case loss in this cohort rather than an end-to-end gain. End-to-end F1 alone can misleadingly suggest that complex policy is “validated” when the benchmark rarely stresses it. Future BAS cohorts should register mechanism-coverage requirements (session-only, endpoint-only, hub over-merge, late horizon, beyond- W_{max} span) before claiming relation-aware gains. M7 intentionally places the final action at 3900 s, outside the frozen $W_{max} = 3600$ s span; production EB4’s zero coverage/F1 is the expected boundary-stress outcome rather than a collection failure.

4.6 Deployed Case-Materialization Trace

We additionally recorded a read-only trace from the deployed investigation surface on 2026-07-23. This trace is artifact-level integration evidence, not another experimental sample and not part of any estimator above. Figure 3(a) shows six workspace-scoped cases, five marked high risk, with both native and NDR-derived sources retained. The selected laboratory case in Figure 3(b) materializes three normalized events—one Suricata NDR event and two Wazuh events—and exposes evidence references for ATT&CK T1018 and T1041. The interface’s 98% operational confidence and `isolate_or_investigate` recommendation belong to the downstream case engine; they are *not* TRUSTEPIISODE’s nullable calibrated P and are not used as evidence of model accuracy. The trace supports only the narrower claim that source-separated evidence and technique support survive into a human-reviewable case view.

4.7 Inclusive Temporal Boundary Fixtures

Unit fixtures on production `EpisodeSynthesizer._candidate` confirm inclusive equality at the frozen contracts: $W_{gap} \in \{299, 300\}$ eligible and 301 ineligible; $W_{max} \in \{3599, 3600\}$ eligible and $\{3601, 3900\}$ ineligible (Table 8). The 3900 s case

Table 4: Primary M1–M7 scenario families (simplified). “Live EB1–EB4 delta?” reports whether fair builders differed on the sealed primary cohort for the intended advanced mechanism.

ID	Ordered activity (summary)	Expected EDR/NDR	Intended mechanism	Live EB1–EB4?
M1	recon $\rightarrow$ local action	process; DNS/flow	entity/time	identical
M2	remote session + SCP archive	process; SSH/SCP flows	session / file_transfer token	identical; token gap
M3	privilege / persistence steps	process; file	causal/entity	identical
M4	lateral-style dual-host steps	process; flows	session/endpoint	identical
M5	staged tooling / execution	process; file/net	entity/time	identical
M6	cleanup / cover tracks	process; file	entity/time	identical
M7	boundary actions at 0/300/3900 s	process; flows	span boundary	boundary failure

Table 5: Primary live M1–M7 cells under production formation (mean $\pm$ sample SD; $n = 5$).

Family	Mode	Coverage	Frag.	Merge	Contam.	F1
M1	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M1	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.300 $\pm$ 0.112	1.000 $\pm$ 0.000
M2	attack-only	0.750 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M2	concurrent-benign	0.750 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.350 $\pm$ 0.091	1.000 $\pm$ 0.000
M3	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M3	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.350 $\pm$ 0.091	1.000 $\pm$ 0.000
M4	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M4	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.367 $\pm$ 0.217	1.000 $\pm$ 0.000
M5	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M5	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.433 $\pm$ 0.091	1.000 $\pm$ 0.000
M6	attack-only	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000
M6	concurrent-benign	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.333 $\pm$ 0.257	1.000 $\pm$ 0.000
M7	attack-only	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	undef.	0.000 $\pm$ 0.000
M7	concurrent-benign	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	undef.	0.000 $\pm$ 0.000

Table 6: Fair sliding-gap baselines versus legacy fixed bins on the sealed 70-run M1–M7 malicious cohort (mean $\pm$ sample SD). Primary comparisons share $W_{gap} = 300$ s and $W_{max} = 3600$ s.

Builder	Role	Coverage	Exact-comp.	Frag.	Contam.	F1
EB0_FIXED_BIN_300s	legacy	0.807 $\pm$ 0.348	0.671 $\pm$ 0.473	0.071 $\pm$ 0.259	0.178 $\pm$ 0.208	0.814 $\pm$ 0.353
EB1_SLIDING_TIME	primary	0.836 $\pm$ 0.335	0.729 $\pm$ 0.448	0.000 $\pm$ 0.000	0.175 $\pm$ 0.207	0.867 $\pm$ 0.338
EB2_SLIDING_ENTITY	primary	0.821 $\pm$ 0.349	0.714 $\pm$ 0.455	0.000 $\pm$ 0.000	0.178 $\pm$ 0.208	0.857 $\pm$ 0.352
EB3_RELATION_NO_ADVANCED_GUARDS	primary	0.836 $\pm$ 0.335	0.729 $\pm$ 0.448	0.000 $\pm$ 0.000	0.175 $\pm$ 0.207	0.863 $\pm$ 0.341
EB4_TRUSTEPIISODE	primary	0.821 $\pm$ 0.349	0.714 $\pm$ 0.455	0.000 $\pm$ 0.000	0.178 $\pm$ 0.208	0.857 $\pm$ 0.352

Table 7: Paired differences EB4–EB1_SLIDING_TIME (contamination/fragmentation flipped so positive favors EB4). Bootstrap: 2000 run-cluster resamples, seed 20260721. Zero deltas reported as no observed difference.

Metric	Mean Δ	Median Δ	95% CI	W/T/L	Interpretation
action_coverage	−0.014	0.000	[−0.043, 0.000]	0/69/1	not resolved
exact_complete_recovery	−0.014	0.000	[−0.043, 0.000]	0/69/1	not resolved
fragmentation	0.000	0.000	[0.000, 0.000]	0/70/0	no observed difference
contamination	0.000	0.000	[0.000, 0.000]	0/60/0	no observed difference
episode_f1	−0.010	0.000	[−0.029, 0.000]	0/69/1	not resolved

Table 8: Inclusive W_{gap}/W_{max} boundary fixtures (unit; all pass).

Dimension	Seconds	Expected	Observed
W_{gap}	299	eligible	eligible
W_{gap}	300	eligible	eligible
W_{gap}	301	ineligible	ineligible
W_{max}	3599	eligible	eligible
W_{max}	3600	eligible	eligible
W_{max}	3601	ineligible	ineligible
W_{max}	3900	ineligible	ineligible

matches the sealed M7 schedule offset and is *not* a substitute for live M7 formation metrics. Results are sealed in results/derived/temporal_boundary_conformance.json.

4.8 Sixty-Run Determinism Matrix

On all 60 sealed malicious runs (M1–M6 $\times$ attack-only/concurrent-benign $\times$ five), D1 canonical ordered replay (10 $\times$ per run; 600 executions) is byte-identical (60/60). D2 separates raw delivery-order robustness (0/60) from re-canonicalized equality (60/60): we do *not*

claim raw-order robustness. D3 ingest-delay grids (including inside, at, and beyond the 900 s revision horizon) pass 60/60; D4 duplicate/idempotent/conflict cases pass 60/60. D5 documents single-writer-per-run ownership: multi-run parallelism is not same-run concurrent mutation, and we claim neither lock-free nor multi-writer determinism.

4.9 Audit Failure Conformance (A1–A12)

RQ2 is supported by the full failure matrix: all seven Assembly-FailureRecord codes (7/7) appear with expected terminal outcomes, plus precedence, exclusivity, idempotent success retry, and append-only rejection (18/18 fixtures pass). This is synthetic forensic audit conformance, not field effectiveness.



(a) Workspace-scoped case queue and selected case summary.



(b) Source-separated evidence cards and ATT&CK support map.

Figure 3: Read-only deployed case trace from the isolated TrustEpisode laboratory. The capture is cropped to remove account-identifying chrome. It demonstrates end-to-end evidence materialization, not calibrated probability or LLM effectiveness.

Table 9: Determinism matrix on all 60 sealed malicious runs (D1–D5).

Test	Runs	Variants	Expected invariant	Pass	Fail	Interpretation
D1 canonical replay	60	10×/run	byte-identical digests	60	0	canonical ordered input
D2 raw delivery order	60	6 perms	equals baseline without reorder	0	60	not order-robust
D2 re-canonicalized	60	6 perms	equals baseline after sort	60	0	canonicalization required
D3 ingest delay	60	horizon grid	successor/late invariants	60	0	prior digest preserved
D4 duplicates	60	conflict/idempotent	reject or no-op	60	0	fail-closed conflicts
D5 single-writer	architecture	ownership	no same-run multi-writer	documented	0	not lock-free concurrency

Table 10: A1–A12 audit conformance fixtures (synthetic; covers all seven AssemblyFailureRecord codes).

Fixture	Expected	Observed	Term.	Prior chg?	Result
A1_SUCCESS	success	success	1	no	pass
A2_SCHEMA_INVALID	schema_invalid	schema_invalid	1	no	pass
A3_KEY_MISMATCH	key_mismatch	key_mismatch	1	no	pass
A4_CONFLICTING_DUPLICATE	conflicting_duplicate	conflicting_duplicate	1	no	pass
A5_DIGEST_MISMATCH	digest_mismatch	digest_mismatch	1	no	pass
A6_VERSION_MISMATCH	version_mismatch	version_mismatch	1	no	pass
A7_MISSING_INPUT	missing_input	missing_input	1	no	pass
A8_ASSEMBLY_TIMEOUT	assembly_timeout	assembly_timeout	1	no	pass
A9_FAILURE_PRECEDENCE_SCHEMA	schema_invalid	schema_invalid	1	no	pass
A9_FAILURE_PRECEDENCE_KEY	key_mismatch	key_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_DUPLICATE	conflicting_duplicate	conflicting_duplicate	1	no	pass
A9_FAILURE_PRECEDENCE_DIGEST	digest_mismatch	digest_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_VERSION	version_mismatch	version_mismatch	1	no	pass
A9_FAILURE_PRECEDENCE_MISSING	missing_input	missing_input	1	no	pass
A9_FAILURE_PRECEDENCE_TIMEOUT	assembly_timeout	assembly_timeout	1	no	pass
A10_TERMINAL_EXCLUSIVITY	rejected	rejected	1	no	pass
A11_IDEMPOTENT_SUCCESS_RETRY	idempotent	idempotent	1	no	pass
A12_APPEND_ONLY_STORE	both_rejected	both_rejected	1	no	pass

Table 11: A–G evidence-loss taxonomy conformance (synthetic A–G plus one sealed live M2 as D).

True class	Predicted	Source	Result
A_execution_failure	A_execution_failure	synthetic	pass
B_sensor_observability_failure	B_sensor_observability_failure	synthetic	pass
C_parser_or_normalization_failure	C_parser_or_normalization_failure	synthetic	pass
D_token_mapping_failure	D_token_mapping_failure	synthetic	pass
E_formation_membership_failure	E_formation_membership_failure	synthetic	pass
F_offline_attribution_failure	F_offline_attribution_failure	synthetic	pass
G_unknown_or_incomplete_trace	G_unknown_or_incomplete_trace	synthetic	pass
D_token_mapping_failure	D_token_mapping_failure	live M2	pass

4.10 Evidence-Loss Taxonomy Validation

The taxonomy is conformance-tested across A–G and instantiated on one sealed live case, M2 (8/8 fixture rows; 1 live). Synthetic passes do not imply field-validated coverage for every class.

4.11 M2 Evidence Waterfall

Final class: D_token_mapping_failure. Action coverage 0.75 with episode F1 1.0 is therefore not evidence that Episode Synthesis dropped an already-normalized file_transfer event. Primary

Table 12: M2 evidence-loss waterfall (10 sealed runs).

Stage	Expected	Observed	Status
CALDERA receipt (linux.scp_archive)	10	10	retained
Raw EDR/NDR observability (scp mentions)	10	10	retained
NormalizedEvent as file_transfer	10	0	drop
Normalized as remote_session instead	10	10	remapped
Episode retains file_transfer	10	0	absent token
Episode retains remote_session	10	10	retained

sealed results are unchanged; any mapper fix requires a new version digest and a post-analysis cohort.

4.12 Contract Ablation Baselines (AB0–AB3)

AB0–AB3 are reference implementations for an isolation study: which contract property prevents which forensic failure mode. They are reference implementations for isolation, not product comparisons.

4.13 Agent-Facing Mutation Conformance

AI0 accepts an unchanged canonical AuditRecord. AI1–AI4 inject a generated fact, leak a label, force an unsupported 0.98 probability, or remove provenance; the closed schema rejects each. AI5–AI6 change a revision or evidence digest without resealing the record; canonical digest verification rejects both. Thus 7/7 cards pass. These are synthetic canonical-record checks, not LLM factuality, deployed authorization, or agent-runtime security tests.

4.14 Held-Out Local-Model Grounding Audits

After a separate unscored 18-card interface smoke test, we froze the system prompt, output schema, four newly valued and re-digested schema-valid AuditRecords, and 20 held-out tasks. We then ran qwen2.5:7b (model digest prefix 845dbda0ea48, Ollama 0.20.5) once with temperature 0 and seed 20260723. Tasks require exact scalar extraction, null/missing-field abstention, or resistance to a task instruction that supplies a false value. The frozen artifact retains prompts, raw responses, token counts, card/record/prompt digests, and per-field scoring.

After observing Qwen, but before downloading or querying guacamole-ai’s default gemma3:1b, we timestamped a locked replication protocol fixing the same prompt, schema, cards, records, seed, and scoring. Gemma (digest prefix 8648f39daa8f) then ran once. Because this replication was registered after the Qwen result, Table 15 is an exploratory transfer audit, not a preregistered or statistical model comparison.

Qwen yields 15/20 exact answers, 15/20 exact citation sets, and 9/20 fully exact responses; one pointer is invalid. Gemma yields only 15/20 schema-shaped responses, 7/20 exact answers, 0/20 exact citation sets, and 0/20 fully exact responses, with 18 invalid pointers. Its raw H17, H18, and H20 responses echo injected numeric or object fragments; this lexical observation was not a frozen metric and is illustrative. The size-sensitive negative result is the point of the boundary: typed input makes failure measurable, but does not make a model accurate. Neither run is field effectiveness, authorization testing, or evidence for general LLM safety.

4.15 Mechanism Conformance Cards

T1–T8 are synthetic. T2/T3 were redesigned so that specific_entity is disabled and session/endpoint becomes the sole legal relation; ablation then fragments. Passes are conformance, not live effectiveness. In the live cohort, EB4 matches EB2 but differs from EB1/EB3 on one concurrent-benign M7 run.

4.16 Supplemental Partition (Not Primary)

Artifacts below live under a separate supplemental_mechanism partition. They do *not* enter Table 5 or the 70-run fair baseline estimator. Evidence classes are stated per row.

- **L1 (live base + offline ablation):** three sealed live M3 SSH cross-host bases; offline ablation links under authenticated_session alone and fragments when that relation is disabled (3/3). Not a claim of organic session-only BAS stress.
- **L2 (offline ablation on L1 seals):** on the same live SSH seals, endpoint_connection alone (entity/session/causal disabled) links denser than ablating endpoint (3/3). Offline-only mechanism check.
- **L3 (offline ablation + controlled inject):** organic hub load is *not* observed (hub-on vs hub-off identical, 0/3 organic). Controlled hub-load inject on the live sealed bases (cutoff+1 peers) makes hub-on vs hub-off diverge (3/3). Controlled inject $\neq$ organic BAS hub stress.
- **L4 late evidence (two distinct classes):** (i) *controlled late inject* on live seals yields successors and beyond_revision_horizon LateEvidenceRecords (3/3); (ii) *wall-clock forwarder hold/release* with real delays: within-horizon 180 s holds seal and show multi-revision successors (2/2 pass); beyond-horizon attempts (930 s hold-release; 1080 s quiet wait) *seal* but do *not* emit LateEvidenceRecords under organic telemetry (2 sealed, 0/2 LateEvidence; new membership formed instead). Beyond-horizon localization is therefore supported by controlled inject, not by wall-clock beyond runs in this lab.

4.17 Artifact Reproducibility

The supplementary archive rebuilds macros from sealed JSON, regenerates tables, and re-runs D1–D5, A1–A12, A–G, AB0–AB3, AI0–AI6, T1–T8, temporal boundary fixtures, and verify_claims.py. It verifies both frozen local-model raw results but does not re-run Ollama. A clean-room extract of the companion ZIP reproduces those checks in the adjacent clean-room report. Digests are refreshed with each packing; stale hashes are not reused.

4.18 Limitations

Two Ubuntu targets; non-commercial EDR; $n = 5$ per cell; controlled benign workloads; no production base rates; fair sliding results largely overlap, but EB4 trails EB1/EB3 on one concurrent-benign M7 run; synthetic cards $\neq$ field validation; raw order not robust; no fitted scorer/calibration; no partial-telemetry/outage; M2 mapping not corrected in the primary cohort. Organic hub stress remains unobserved on live streams. Wall-clock *within*-horizon holds are executed (2/2); wall-clock *beyond*-horizon LateEvidenceRecord under organic telemetry was attempted (2 sealed) but not observed.

Table 13: Contract ablation / reference-implementation comparison (synthetic isolation study; not SOTA product baselines).

Property	AB0 Mutable	AB1 Noncan.	AB2 Silent	TrustEpisode	Evidence
prior_history_immutable_under_late	no	no	yes	yes	synthetic conformance
byte_identical_replay_under_ref_reorder	yes	no	yes	yes	synthetic conformance
explicit_terminal_failure_on_missing_input	no	no	no	yes	synthetic conformance
append_only_terminal_store	no	yes	yes	yes	synthetic conformance

Table 14: Downstream-agent canonical-record mutation cards (synthetic conformance only; not LLM accuracy or deployed authorization).

Card	Operation	Guard	Expected	Observed	Result
AI0	consume unchanged canonical AuditRecord	closed schema plus canonical digest	accept	accept	pass
AI1	add agent_asserted_fact to canonical record	additionalProperties false	schema reject	schema reject	pass
AI2	add malicious_label to online record	additionalProperties false	schema reject	schema reject	pass
AI3	replace out-of-support null P with 0.98	support-probability coupling	schema reject	schema reject	pass
AI4	remove ordered evidence references	required provenance field	schema reject	schema reject	pass
AI5	swap revision digest without resealing record	canonical digest binding	digest reject	digest reject	pass
AI6	swap evidence content digest without resealing record	canonical digest binding	digest reject	digest reject	pass

Table 15: Held-out AuditRecord pilots. Gemma was locked after the Qwen result but before model download. Exact requires answer, abstention flag, and citation set to match.

Model	JSON	Answer	Abst.	Cite	Exact	Bad ptr.
Qwen2.5-7B	20/20	15/20	14/20	15/20	9/20	1
Gemma3-1B	15/20	7/20	13/20	0/20	0/20	18

L3 hub divergence and L4 beyond-horizon localization use controlled inject on live seals where noted. M7 is primary boundary stress and should not be interpreted as successful 3900 s recovery under a 3600 s span. AI0–AI6 test canonical-record validation, not deployed service-identity enforcement. The LLM audits are synthetic, two-model, same-card, and once-run; the Gemma replication was locked only after Qwen was observed. Their 9/20 and 0/20 exact rates cannot establish general factuality, a size effect, robustness, or operational utility.

Analysis status. Complete (primary): live M1–M7 reconstruction (110 sealed train runs with B1–B8); fair sliding baselines with paired CI; T1–T8; M2 waterfall; 60-run determinism D1–D5; A1–A12; A–G; AB0–AB3; AI0–AI6; held-out Qwen audit + locked Gemma replication; inclusive boundary fixtures including 3900 s. Complete (supplemental, not primary): L1 live bases + offline L1/L2; L3 controlled hub inject; L4 controlled late inject; wall-clock within 2/2 and beyond 2 sealed/0 LateEvidence. Not executed: organic hub-stressed BAS traffic; fitted scorer / RP / PO1.

5 Conclusion

TRUSTEPISODE provides a deterministic, auditable EDR/NDR episode-reconstruction contract and a read-only evidence boundary for AI-assisted investigation: single-owner immutable revisions, append-only terminal outcomes with a complete failure-code matrix, and stage-localized A–G evidence-loss diagnosis. On the sealed M1–M7 cohort, sliding sessionization improves completeness relative to fixed bins, but fair sliding time/entity/relation builders do not establish TrustEpisode superiority. EB4 matches EB2 on all 70 action-coverage pairs, but trails EB1/EB3 on one concurrent-benign M7 run; the EB4–EB1 action-coverage estimate is $\Delta = -0.014$ with 95% CI $[-0.043, 0.000]$. That mostly-overlapping result is forensic evidence about benchmark coverage and one boundary-case cost: advanced guards require targeted conformance cards, which we provide without treating them as live effectiveness. Canonical replay is byte-identical on all 60 sealed malicious runs;

raw arrival-order robustness requires re-canonicalization. M2’s missing file_transfer token is D_token_mapping_failure, not a demonstrated formation drop. Contract ablation shows which properties prevent mutable history, silent missing-input failure, and noncanonical replay. Seven downstream mutation cards accept the valid canonical control and reject injected facts, label leakage, unsupported probability, removed provenance, and unsealed digest tampering; they do not evaluate deployed authorization. A separate once-run held-out Qwen2.5-7B pilot yields 15/20 exact answers but only 9/20 fully exact answer/abstention/citation triples. A same-card replication, locked after that result but before querying deployment-default Gemma3-1B, yields 7/20 exact answers, no exact citation sets, no fully exact responses, and 18 invalid pointers. This is evidence that a typed record makes size-sensitive model failure inspectable, not that either model is generally accurate or that the observed gap estimates a size effect. Supplemental partition artifacts (not primary estimators) include L1 live bases with offline L1/L2 ablations, L3 controlled hub inject, L4 controlled late inject, wall-clock within-horizon holds (2/2), and sealed beyond-horizon attempts without organic LateEvidenceRecord. Sealed M7 is a primary 3900 s boundary-stress family with expected zero recovery beyond W_{max} . Fitted scoring, calibration, and telemetry-outage robustness remain unevaluated and outside these claims. A deployed read-only trace confirms that source-separated Suricata/Wazuh evidence and ATT&CK references reach the analyst case view; its displayed confidence is not treated as the contract’s probability or as LLM-effectiveness evidence. The result is a falsifiable evidence substrate for future AI reasoning; both negative audits reinforce rather than remove the need to distrust the model.

A Frozen Submission Contracts

A.1 Machine Schema and Audit Assembly

The normative schema-ready in-PDF contract follows JSON Schema Draft 2020-12 closed-object semantics [22] for NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, FeatureVector, SupportDecision, AuditRecord, and AssemblyFailureRecord. FeatureVector owns D, C, m, o^{det} without health input. SupportDecision owns resolved a^{det}, κ, g , detector count n , eligibility reason, decision_eligible, EDR/NDR coverage_ref_ids, and health_snapshot_digest. Count $n = 0$ forces out-of-support, false eligibility, and null P even if both detectors report

Table 16: Targeted synthetic mechanism cards (conformance only; not mixed into primary effectiveness).

Card	Result	Expected	Observed
T1-PROCESS-CAUSAL	pass	fragment without causal	full-eps:1, abl-eps:2
T2-AUTH-SESSION	pass	session only links ablation fragm...	session-only:1, abl:2
T3-ENDPOINT-MAPPING	pass	endpoint only links ablation frag...	endpoint-only:1, abl:2
T4-HUB-GUARD	pass	hub guard changes membership or e...	full-eps:4, no-hub-eps:1
T5-PARENT-MERGE	pass	merge enabled creates parent epis...	full-parents:3, abl-parents:0
T6-LATE-SUCCESSOR	pass	disabling late successor increase...	full-rev:4, abl-rev:3, full-late:...
T7-BEYOND-HORIZON	pass	beyond horizon emits LateEvidence...	late:1, reasons:(beyond-revision-...
T8-TIE-BREAK	pass	canonical arrival order yields id...	equal:True; raw-reverse-differs:True

no_detection. AuditRecord requires those outputs, z, P, U , the revision digest, health snapshot, ordered references, exact versions, and canonical digest; out-of-support requires JSON null P and false eligibility. For each revision key the Assembler atomically inserts one terminal outcome: AuditRecord after a complete exact-version join, or AssemblyFailureRecord after conflict or deadline. The assembly deadline is revision.watermark + 120 s. The unique-key store is an insert/select sink and cannot later replace failure with success.

Identifiers are ASCII and sort by unsigned UTF-8 bytes. AuditRecord uses string IDs/digests, nonnegative integer revision, finite $D, P \in [0, 1]$, $C \in \{0, 1\}$, numeric z , Boolean eligibility, closed enums/maps for κ, U, m, h, a^{det} , ordered reference arrays, and a closed version object. Only P and group key are nullable: out-of-support requires $P = \text{JSON null}$; only supported-group requires a non-null key. NaN, infinities, negative zero, and binary64 overflow are invalid. AssemblyFailureRecord contains no D, C, z, P, U and has exactly one of the seven failure codes defined by the assembly contract (schema, key, duplicate, digest, version, missing-input, timeout). Canonical bytes follow RFC 8785 JSON canonicalization [13]. JCS sorts object properties but preserves array order, so the application sorts every reference array first. SHA-256 includes each outcome’s declared digest-field set except canonical_digest. AuditRecord includes type/schema/key/revision, revision digest, $D, C, z, P, \kappa, g, e, U, m, h, o^{det}, a^{det}$, detector count/reason, ordered references, and versions; AssemblyFailureRecord includes type/schema/key/revision, revision digest, failure code, observed input types, and assembly-policy version; LateEvidenceRecord includes type/schema/episode/latest revision and digest/event, arrival/deadline/reason, raw pointer/root, ordered lineage references, and VersionSet. Host, PID, wall time, and duration are an optional non-canonical envelope. References sort by relation rank, dependency-group bytes, object-before-sentinel, object-ID-or-empty bytes, then digest-or-empty bytes. Optional absence uses a closed sentinel; required absence fails assembly. Assembly input order is revision, feature, logit, support, probability, U , then coverage. Failure precedence is schema, key, duplicate, digest, version, missing, then timeout. Every VersionSet component closes both a version identifier and content digest for schema, parser, formation, detector registry, feature, normalizer, model, calibrator, support, sufficiency, assembly, and manifest. Conforming validation must accept supported, unsupported, late, and assembly-failure records and reject any online record containing malicious_label. Online objects reject all label, scenario, marker/window, and ability/operation fields.

A.2 Episode Algorithm and State Machine

Fixed values are $W_{gap} = 300$ s, $W_{max} = 3600$ s, $L_{wm} = 120$ s, $W_{rev} = 900$ s, and $W_{ret} = 86400$ s. Relations have $\rho = (4, 3, 2, 1)$ for causal/session/endpoint/entity and n_R counts satisfied classes; candidates sort by $(-\rho, -n_R, \Delta_t, t_{start}, id)$. A hub has at least 25 relations in 600 s. Merge permits at most three parents connected by causal, session, or endpoint relations and at most 500 events/200 entities; otherwise attach to the first candidate. The event order is $(t^{event}, t^{ingest}, source_instance, event_id)$. First finalization freezes t_f and deadline $t_f + 900$ s. The lineage then leaves the open index and remains in a revisable index; at closure, canonical relation keys remain in a closed-lineage index through deadline plus W_{ret} . Each event queries revisable/closed indexes first and returns after exactly one successor/LateEvidenceRecord; only no match permits open start/attach/merge. A qualifying late event before the fixed deadline creates a finalized successor without reopening or extending the horizon; the deadline creates a membership-identical closed successor. A later related event creates only a LateEvidenceRecord in the late store with reason beyond_revision_horizon, closed_lineage_match, or duplicate_after_terminal_outcome. Replay equality covers every state, late record, output, reference, and digest.

A.3 Thresholds and Sufficiency Rules

Table 17 is the single numeric and precedence registry for the submission PDF.

Feature Extractor emits only detected, examined_none, or not_examined. At EpisodeRevision.event_time_end, Scope Resolver combines that observation with exactly one content-distinct covering SourceCoverage interval per expected instance and emits available, no_detection, collector_unavailable, parser_invalid, or unknown; the first two are health-support states, but zero eligible detector values still force out-of-support. CR1 is an overlapping endpoint-to-host binding conflict, CR2 is canonical EDR/NDR flow-direction conflict, and CR3 is conflicting parent/root lineage for one process key. Health is healthy for zero missed 30 s heartbeats, lag ≤ 120 s, valid parser, and live transport; degraded for one missed heartbeat or lag (120, 600] s; and down for two misses or authenticated stop. Required provenance is every membership/root, selected link, scored input, coverage object, and version; diagnostics are optional. Any later executable threshold file must exactly match Table 17.

Table 17: Frozen numeric registry and deterministic precedence.

Contract owner	Parameter(s)	Frozen value / order	Validation rule
A. Numeric contract			
Formation	$W_{gap}; W_{max}; L_{wm}; W_{rev}; W_{ret}$	300; 3600; 120; 900; 86400 s	sealed before final training fit
Formation guards	hub relations/window; parents; component cap	25/600 s; 3; 500 events/200 entities	reject on violation; never learned
Feature	detector IDs; top- k ; source mask	3 EDR + 3 NDR; $k = 3$; EDR+NDR	finite [0, 1]; mid-rank CDF
Offline matching	$\theta_{mal}; \theta_{contam}; \theta_{match}$	.50; .25; .50	freeze before label release
Match predicates	temporal IoU; entity Jaccard	.10; .50	also require exact action/receipt
Uncertainty (paired CI)	bootstrap repetitions; seed	2000; 20260721	resample complete run clusters
B. Deterministic U precedence (first match wins)			
$u_{coverage}$	health/parser state	unknown > unavailable > degraded > complete	evaluate authenticated coverage first
$u_{provenance}$	required references	missing > complete	partial reserved; v1 emits only missing/complete
$u_{lateness}$	revision state	late_accepted > revisable > stable	accepted successor; before deadline; closed
$u_{contradiction}$	CR1–CR3	present > unevaluable > none	first applicable conflict rule wins
$u_{support}$	exact support enum	out_of_support; supported_group; supported_global	design-only; no fitted P claimed

A.4 Probability Interface Contract (Unevaluated)

Optional score fields may remain in schema. Binding rules only: version identifiers and digests must match; unsupported probability is JSON null; U must not rewrite P ; no fitted coefficients are reported in this submission. Affine objectives, group gates, fold assignment, and refit protocols are deferred to supplementary material and are not empirical contributions here.

A.5 Live-System Lab, Cards, and Offline Labels

The lab, card, and label contract is exactly Sections 4.2–4.3 and Equation 1. It pins public CALDERA and OCSF versions; missing manifest fields fail closed. The amended primary cohort fixes five valid runs per M1–M7 card/mode and five standalone runs per B1–B8 card, all achieved. Each Linux-only M/B card closes image key and digest, ordered ability IDs, no-profile Bash executor plus exact command templates, monotonic offsets, fixed target paths, offline receipt preimage, success/failure predicates, cleanup commands and digest check, and allowed mode. Target commands cannot receive run/card/ability/operation/receipt identifiers or values; orchestrator-only receipts remain in Ground Truth Store. Template variables resolve only from the frozen run manifest. Malicious modes are attack-only or one hash-selected B1–B8 workload begun 60 s earlier; benign cards also run standalone. Missing values and malformed inputs are counted, not zero-filled.

A.6 Unexecuted Extensions and Statistical Contract

Formation coverage, fragmentation, over-merge, exact-complete recovery, and one-to-one episode F1 are reported for the amended sealed cohort. Fitted calibration, locked-test ranking, RP0–RP7 partial-telemetry children, physical outage, and full assembly cost benchmarks remain registered but unexecuted. The pre-declared group-calibrator gate (50 positive, 50 negative, 10 runs) is not lowered post hoc. If those protocols are later executed, resampling units must be complete runs; undefined and failed outcomes must be counted, never silently zeroed; and protocol amendments must be timestamped before inspecting estimates.

The required supplementary archive is TrustEpisode_reproducibility_companion_v1.zip. Concatenating the following two hexadecimal lines gives its SHA-256:

```
91da6f143bc4c5a137cf3b8d2584c3c6
476c79950d3d481b67164f25d51ae046
```

The adjacent sidecar records the same digest.

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